Categorical data analysis_GEE

McNemar test & GEE

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McNemar test

```r
path = "http://www.stat.ufl.edu/~aa/cat/data/"
Opinions <- read.table(paste0(c(path,"Envir_opinions.dat"),
                              collapse = ""), header=TRUE)
tab <- xtabs(~y1 + y2, data = Opinions)
tab
```

```
   y2
y1    1   2
  1 227 132
  2 107 678
```

```r
mcnemar.test(tab, correct=FALSE)
```

```

    McNemar's Chi-squared test

data:  tab
McNemar's chi-squared = 2.6151, df = 1, p-value = 0.1059
```

Wald confidence interval

```r
#install.packages("PropCIs")
library(PropCIs)
diffpropci.Wald.mp(107, 132, 1144, 0.95)
```

```
data:

95 percent confidence interval:
 -0.004602847  0.048309140
sample estimates:
[1] 0.02185315
```

Matched pairs estimation

```r
#install.packages("gee")
library(gee)
```

```
Warning: package 'gee' was built under R version 4.3.3
```

```r
Opinions <- read.table(paste0(c(path,"Opinions.dat"),
                              collapse = ""),header=TRUE)

try <- glm(y ~ question, family = binomial,data = Opinions)
summary(try)
```

```
Call:
glm(formula = y ~ question, family = binomial, data = Opinions)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.88589    0.06503 -13.623   <2e-16 ***
question     0.10353    0.09104   1.137    0.255
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2806.4  on 2287  degrees of freedom
Residual deviance: 2805.1  on 2286  degrees of freedom
AIC: 2809.1

Number of Fisher Scoring iterations: 4
```

```r
try$coefficients
```

```
(Intercept)    question
 -0.8858933   0.1035319
```

```r
fit <- gee(y ~ question, id=person, family=binomial(link="logit"),
           corstr="independence", data=Opinions)
```

```
Beginning Cgee S-function, @(#) geeformula.q 4.13 98/01/27

running glm to get initial regression estimate

(Intercept)    question
 -0.8858933   0.1035319
```

```r
# id identifies variable on which observe y1, y2
summary(fit)
```

```

 GEE:  GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
 gee S-function, version 4.13 modified 98/01/27 (1998)

Model:
 Link:                      Logit
 Variance to Mean Relation: Binomial
 Correlation Structure:     Independent

Call:
gee(formula = y ~ question, id = person, data = Opinions, family = binomial(link = "logit"),
    corstr = "independence")

Summary of Residuals:
       Min         1Q     Median         3Q        Max
-0.3138112 -0.3138112 -0.2919580  0.6861888  0.7080420


Coefficients:
              Estimate Naive S.E.    Naive z Robust S.E.   Robust z
(Intercept) -0.8858933 0.06505597 -13.617401  0.06502753 -13.623357
question     0.1035319 0.09107816   1.136737  0.06397794   1.618244

Estimated Scale Parameter:  1.000875
Number of Iterations:  1

Working Correlation
```

```
     [,1] [,2]
[1,]    1    0
[2,]    0    1
```

```r
fit$coefficients
```

```
(Intercept)    question
 -0.8858933   0.1035319
```

```r
fit2 <- gee(y ~ question, id=person, family=binomial(link=logit),
            data=Opinions)
```

```
Beginning Cgee S-function, @(#) geeformula.q 4.13 98/01/27
running glm to get initial regression estimate

(Intercept)    question
 -0.8858933   0.1035319
```

```r
summary(fit2)
```

```
 GEE:  GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
 gee S-function, version 4.13 modified 98/01/27 (1998)

Model:
 Link:                      Logit
 Variance to Mean Relation: Binomial
 Correlation Structure:     Independent

Call:
gee(formula = y ~ question, id = person, data = Opinions, family = binomial(link = logit))

Summary of Residuals:
       Min         1Q     Median         3Q        Max
-0.3138112 -0.3138112 -0.2919580  0.6861888  0.7080420


Coefficients:
              Estimate Naive S.E.    Naive z Robust S.E.   Robust z
(Intercept) -0.8858933 0.06505597 -13.617401  0.06502753 -13.623357
```

```
question     0.1035319 0.09107816   1.136737  0.06397794   1.618244

Estimated Scale Parameter:  1.000875
Number of Iterations:  1

Working Correlation
     [,1] [,2]
[1,]    1    0
[2,]    0    1
```

GEE with different working correlation matrix

```
Abortion <- read.table(paste0(c(path,"Abortion.dat"),
                              collapse = ""),header=TRUE)

sit <- factor(Abortion$situation, levels=c(3,1,2))
```

Assuming data are independent

```
fit.glm <- glm(response ~ sit + gender, family=binomial,
               data=Abortion)
summary(fit.glm) # ML estimates for 5550 independent observations
```

```
Call:
glm(formula = response ~ sit + gender, family = binomial, data = Abortion)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.125408   0.055601  -2.255   0.0241 *
sit1         0.149347   0.065825   2.269   0.0233 *
sit2         0.052018   0.065843   0.790   0.4295
gender       0.003582   0.054138   0.066   0.9472
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7689.5  on 5549  degrees of freedom
Residual deviance: 7684.2  on 5546  degrees of freedom
AIC: 7692.2

Number of Fisher Scoring iterations: 3
```

Assuming correlation structure : identity(independence)

```r
fit.gee <- gee(response ~ sit + gender, id=person, family=binomial,
                 corstr="independence", data=Abortion)
```

```
Beginning Cgee S-function, @(#) geeformula.q 4.13 98/01/27

running glm to get initial regression estimate

 (Intercept)         sit1         sit2       gender
-0.125407576  0.149347113  0.052017989  0.003582051
```

```r
# cluster on "id" variable
summary(fit.gee)
```

```

 GEE:  GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
 gee S-function, version 4.13 modified 98/01/27 (1998)

Model:
 Link:                      Logit
 Variance to Mean Relation: Binomial
 Correlation Structure:     Independent

Call:
gee(formula = response ~ sit + gender, id = person, data = Abortion,
    family = binomial, corstr = "independence")

Summary of Residuals:
       Min         1Q     Median         3Q        Max
-0.5068800 -0.4825552 -0.4686891  0.5174448  0.5313109


Coefficients:
                Estimate Naive S.E.     Naive z Robust S.E.    Robust z
(Intercept) -0.125407576 0.05562131 -2.25466795  0.06758236 -1.85562596
sit1         0.149347113 0.06584875  2.26803253  0.02973865  5.02198759
sit2         0.052017989 0.06586692  0.78974374  0.02704704  1.92324166
gender       0.003582051 0.05415761  0.06614123  0.08784012  0.04077921

Estimated Scale Parameter:  1.000721
Number of Iterations:  1
```

```
Working Correlation
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    1    0
[3,]    0    0    1
```

Assuming correlation structure : exchangeable

```r
fit.gee2 <- gee(response ~ sit + gender, id=person, family=binomial,
                corstr="exchangeable", data=Abortion)
```

```
Beginning Cgee S-function, @(#) geeformula.q 4.13 98/01/27
```

```
running glm to get initial regression estimate
```

```
 (Intercept)         sit1         sit2       gender 
-0.125407576  0.149347113  0.052017989  0.003582051 
```

```r
summary(fit.gee2)
```

```

 GEE:  GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
 gee S-function, version 4.13 modified 98/01/27 (1998) 

Model:
 Link:                      Logit 
 Variance to Mean Relation: Binomial 
 Correlation Structure:     Exchangeable 

Call:
gee(formula = response ~ sit + gender, id = person, data = Abortion, 
    family = binomial, corstr = "exchangeable")

Summary of Residuals:
       Min         1Q     Median         3Q        Max 
-0.5068644 -0.4825396 -0.4687095  0.5174604  0.5312905 
```

```
Coefficients:
                Estimate Naive S.E.     Naive z Robust S.E.    Robust z
(Intercept) -0.125325730 0.06782579 -1.84775925  0.06758212 -1.85442135
sit1         0.149347107 0.02814374  5.30658404  0.02973865  5.02198729
sit2         0.052017986 0.02815145  1.84779075  0.02704703  1.92324179
gender       0.003437873 0.08790630  0.03910838  0.08784072  0.03913758

Estimated Scale Parameter:  1.000721
Number of Iterations:  2

Working Correlation
          [,1]      [,2]      [,3]
[1,] 1.0000000 0.8173308 0.8173308
[2,] 0.8173308 1.0000000 0.8173308
[3,] 0.8173308 0.8173308 1.0000000
```

```r
table(Abortion$gender)
```

```
   0    1
2439 3111
```